

AMR YASSER

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EDUCATION

University of Science and Technology, Zewail City

Sep 2023 – Expected Jun 2027

B.Sc. in Computational Science and AI, Concentration: Data Science and AI

Giza, Egypt

- **Relevant coursework:** Machine Learning, Deep Learning, Computer Vision, Statistical Inference, Linear & Non-linear Programming, Probability & Statistics, Linear Algebra, Algorithm Design & Analysis, Software Engineering, Information Theory.

PROJECTS

- **Network Intrusion Detection — Kaggle Competition (2nd of 40+ teams):** Built a multi-class classifier (benign, DDoS, DoS, Mirai, Recon, MITM) on imbalanced network telemetry. Implemented leakage-safe preprocessing with SMOTE applied only to training folds, domain-specific feature engineering, and a three-stage feature selection pipeline. Trained GPU-accelerated XGBoost and LightGBM with Optuna hyperparameter tuning, stacked into a logistic-regression meta-learner. **Private score: 0.9163.**
- **Adaptive Code Lab — Automated Code Grading Platform:** Built a full-stack educational platform with strict Clean Architecture (domain, application, infrastructure, web layers). Integrated Google Gemini for plagiarism-detection embeddings and Groq's Llama-3.3-70B for AI-generated hints. Sandboxed code execution via the Piston API. Achieved 97.52% test coverage. Stack: Python, Flask, SQLite, pytest.
- **Revised Simplex Method — LP Solver from Scratch:** Implemented the Revised Simplex Method for linear programming including LU-decomposition basis updates, Bland's rule for cycling prevention, and Phase-I/Phase-II handling. Benchmarked against scipy's HiGHS solver on Klee-Minty cubes, dense random LPs, and sparse problems.
- **Convolutional AE & β -VAE on Medical MNIST:** Trained a convolutional autoencoder and a β -Variational Autoencoder on PathMNIST in TensorFlow/Keras. Diagnosed posterior collapse during a $\beta=1$ pilot run and resolved it by tuning to $\beta=0.25$, stabilizing KL divergence at ~ 43 nats. Modular source package with 20 unit tests. Test MSE: 0.00429 (AE), 0.00699 (VAE).
- **Video Keyframe Summarization (BiLSTM vs. Transformer):** Built dual-architecture sequence regression models for video summarization on TVSum/SumMe datasets using frozen MobileNetV3-Large feature extraction. Implemented a custom Interpretable Transformer with attention map extraction and zero-shot transfer evaluation.

WORK EXPERIENCE

Zewail City University

Spring 2026 (ongoing)

Junior Teaching Assistant — Deep Learning (DSAI 309)

Giza, Egypt

- Supporting course staff and student cohort: facilitating communication between TAs, students, and the course instructor; running supplementary sessions and one-on-one tutoring on deep learning fundamentals and PyTorch implementation.

Zewail City University

Jan 2024 – May 2025

Junior Teaching Assistant — Linear Algebra & Vector Geometry (MATH 104)[†]

Giza, Egypt

- Created summary materials and ran supplementary online sessions and one-on-one tutoring for students within a 300-person cohort; collaborated with senior TAs to address common conceptual gaps.

LEADERSHIP & ACTIVITIES

AWS Cloud Club — Zewail City

2025 – Present

HR Head

Giza, Egypt

- Leading the HR function for the student branch: member recruitment, team coordination, and internal communications.

IEEE Zewail City Student Branch

2024 – 2025

HR Vice Head[†]

Giza, Egypt

- Recognized for dedication and contributions throughout the 2024/2025 season (Certificate of Recognition, signed by IEEE ZC President and Supervisor).

CERTIFICATIONS

- **NVIDIA — Building LLM Applications with Prompt Engineering[†]** (Certificate of Competency, Sept 2025).
- **NVIDIA — AI for All: From Basics to GenAI Practice[†]** (Sept 2025).

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, C#, R, SQL, JavaScript
- **ML & Scientific Libraries:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, NumPy, SciPy, Pandas, Matplotlib, OpenCV, Optuna
- **Frameworks & Tools:** Flask, FastAPI, pytest, Git, Docker, Linux/Bash, SQLite, LaTeX
- **LLM Integration:** Google Gemini API, Groq API, prompt engineering